

# Hybrid Biologically-Informed Neural Network Validates the Exclusion of Unmapped Genes in Cancer Survival Analysis

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## Abstract

Biologically-informed neural networks (BINNs) have emerged as performant, interpretable models that incorporate biological priors into model architecture. They can be used to map biological data, such as gene expression data, to outputs of clinical interest, such as survival predictions. However, BINNs rely on gene-pathway mappings from biological knowledge bases. Yet a large portion of genes in sequencing data are unmapped. We introduce a hybrid BINN model, HyBINN, to enable a rigorous ablation study to test whether these unmapped genes carry additional predictive power. Using data from a TCGA breast cancer cohort, we perform an ablation study, benchmark HyBINN against classical models, and perform pathway interpretability to gain further insights into the limitations of BINNs in a high dimension, low sample size (HDLSS) single-omic data context. Our results validate the exclusion of unmapped genes common in BINN work, but also demonstrate the failure modes of BINNs in this limited data regime.

***Index Terms*** — breast cancer, survival analysis, biologically-informed neural network, gene expression

## 1. INTRODUCTION

Breast cancer is a heterogeneous disease with great variability in clinical outcomes. Predicting the survival of cancer patients is a key step for informing treatment and enabling precision medicine. Increasingly, high-dimensional omics data obtained by sequencing individual patient tumors is being used to inform survival predictions. This data provides high levels of quantitative detail into tumor biology compared with traditional T/N/M staging by a clinician. Machine learning, particularly deep learning (DL), has advantages over traditional methods for handling this complex data. As such, DL has been extensively applied to cancer survival analysis in recent years [1].

When building models for clinical decision support, interpretability is critical. Neural networks are “black boxes” — their decision process cannot be adequately understood nor explained due to model complexity. To address this, Biologically-Informed Neural Networks (BINNs) base their architecture off biological knowledge such as the mapping of genes to pathways. These biological priors provide ante-hoc interpretability and may also improve generalization on challenging omics data modalities such as gene expression [2], [3].

In the context of genomic data, BINNs use gene-pathway mappings to create a semantically meaningful sparsely-connected layer. These mappings, obtained from knowledge bases such as

Reactome [4], are limited to our current understanding of biology and only contain certain genes. Unmapped genes — that is, the genes not present in knowledge bases — are usually discarded and remain understudied in a BINN context. Do these genes carry additional predictive power, or is it valid to discard them as noise? By building a hybrid sparse-dense model that considers both mapped and unmapped genes, we present a rigorous ablation study and an honest null result testing the assumption that unmapped genes can be ignored in favor of purely pathway-based models.

## 2. BACKGROUND

Omics data is often high-dimension low-sample size (HDLSS) — small cohorts of patients and massive feature spaces (e.g. tens of thousands of genes). This presents challenges for survival analysis models: the curse of dimensionality causes data sparsity, overfitting, and spurious correlations. A variety of techniques such as regularization, dimensionality reduction, cross-validation, and others can improve performance on HDLSS data. In the context of omics data, structural inductive biases based on biological priors is an especially effective technique.

Survival analysis can be modeled as a time-to-event problem. The most common traditional model for survival analysis is the Cox Proportional Hazards (Cox PH) model [5], which considers the effect of covariates on survival. However, it is a linear model that may struggle to model complex relationships or high-dimensional data. To this end, modern research has leveraged neural networks to better capture these relationships between covariates for survival analysis. Deep survival networks such as DeepSurv [6] and DeepHit [7] have demonstrated superior performance to Cox models.

BINNs have shown particular promise for integrating HDLSS omics data into survival analysis. These models can handle genomic data by encoding inductive bias into model structure. Nodes in the input layer represent genes, while nodes in the first hidden layer represent pathways. Sparse connections are enforced via a gene-pathway mask matrix, where a connection between gene  $i$  and pathway  $j$  only exists if gene  $i$  participates in pathway  $j$ . Rather than random dropout or regularization, this type of sparsity guides the model using biological knowledge.

Key pathway-informed models demonstrating these advantages include Cox-PASNet [8] and its successor PAGE-Net [9], P-Net [10] for prostate cancer, PathExpSurv [11], and other multi-omic models [12], [13]. These models have demonstrated superior performance to Cox methods and simple dense neural networks, though standardized performance comparison is currently lacking in the field [2]. However, all of them have only considered mapped genes. PathExpSurv allowed input genes to associate with new pathways after pre-training on the sparse BINN, but still did not appear to consider unmapped genes.

There are a few major knowledge bases for gene-pathway mapping, notably Reactome, KEGG, and MSigDB. For this study, we consider Reactome [4]. Coverage and quality may vary, but they all share the limitation of covering only a subset of the human genome. While major oncogenes are more easily identifiable, it is reasonable to assume there are still many relevant gene-pathway interactions that have yet to be annotated. Thus, the unmapped genes may carry additional predictive power. So while pathway-aware BINNs have shown great success, they assume that

unmapped genes are irrelevant and discard them. This assumption should be tested to validate the use of BINNs for genomic survival analysis.

### 3. METHODS

#### Dataset

We used data from The Cancer Genome Atlas (TCGA) Breast Cancer (BRCA), downloaded from the UCSC Xena portal [14]. The STAR TPM Gene Expression RNAseq, Phenotype, and Survival datasets were used for genomic data, clinical variables, and survival labels, respectively. The raw gene dataset contained 1,226 samples and 60,661 gene features. The raw phenotype dataset contained 1,255 samples and 86 clinical features.

#### Preprocessing

The gene dataset was rigorously preprocessed to reduce the feature space. It was filtered to only use primary tumor samples, remove PAR\_Y genes, only use protein-coding genes (via GENCODE human gene annotation table v23), remove low-expression and low-variance genes, and standardize values to a standard normal distribution ( $\mu=0$  and  $\sigma=1$ ). After filtering, the feature space was reduced to 11,386 genes.

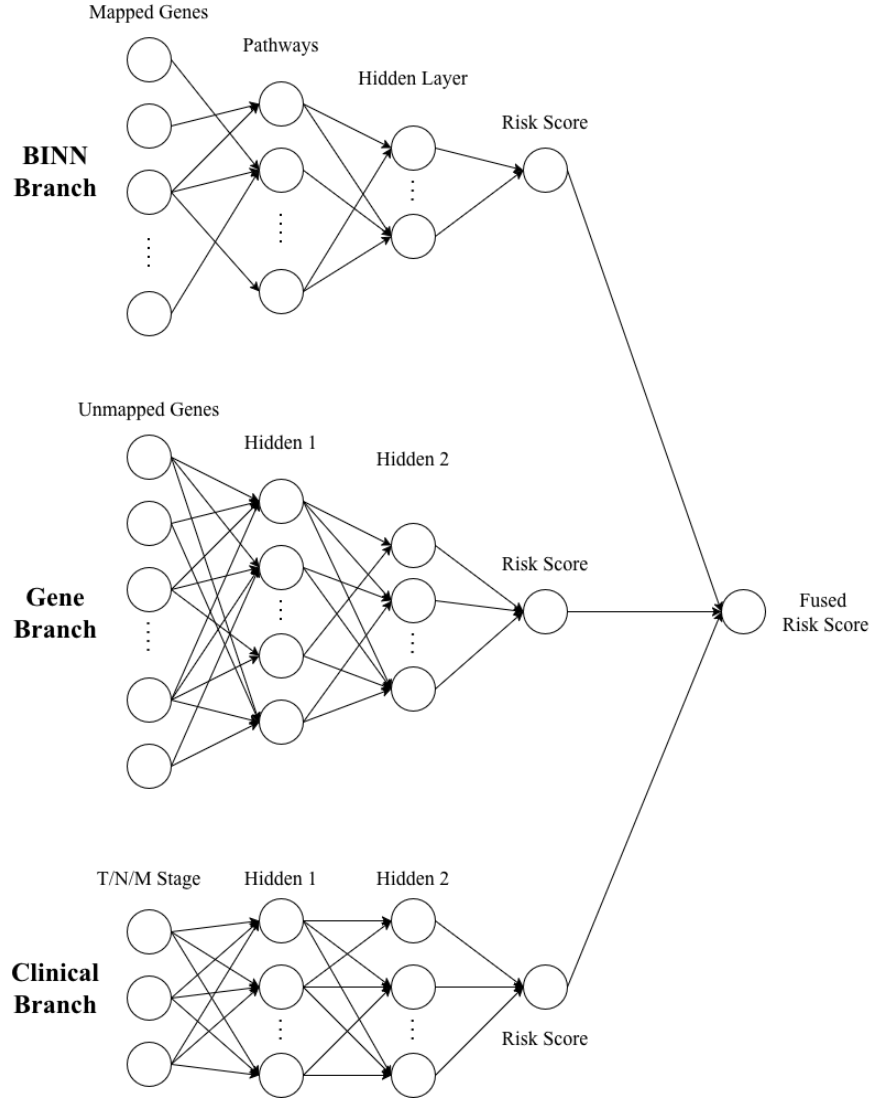
The clinical data also underwent defensible feature selection. It was filtered to only use primary tumor samples, drop mostly-null features, drop no- and low-variance features, remove administrative/ID features, drop spurious features or features that leak survival outcome, and collapse redundant features. Of the remaining features, race, age, synchronous malignancy, and primary diagnosis were dropped since they had low significance when fitted with a simple Cox

PH model. Ethnicity was dropped due to its large class imbalance and different context from genomics. Finally, only the T, N, and M prognostic stages remained. They were integer encoded and aligned with the gene and survival data.

After filtering and inner joining the three datasets, 1,059 patients remained. The dataset contains severe class imbalance for the event outcome, with  $n=911$  patients being censored ( $OS=0$ ) and  $n=148$  patients deceased ( $OS=1$ ).

### Model Design

We have constructed HyBINN, a feed-forward neural network with three input branches (Figure 1).



**Figure 1. HyBINN Model Architecture.** The model contains three branches.

The model splits gene expression data into mapped and unmapped genes based on a Reactome pathway mapping file. Mapped genes are routed to the BINN branch, with pathway-enforced sparsity in the first layer. With initial pathway weights  $W$  and gene-pathway mask matrix  $M$ , the masked weights  $W_m$  are calculated as:

$$W_m = WM^T$$

The hidden layer is intended to capture biological processes and hierarchical relationships and simply uses dropout-enforced sparsity. Tanh activation functions are used on all nodes, consistent with Cox-PASNet [8].

Unmapped genes are routed to the gene branch: a standard MLP with dropout, BatchNorm, and ReLU activations on all nodes. The three chosen clinical variables are processed through the clinical branch: a small MLP with two hidden layers and light dropout. Each branch independently outputs a single risk score, with no activation or bias.

The final fused risk score is simply a linear combination of the risk scores of any active branches multiplied by population-level weights. With  $n$  branches active:

$$r_{final} = \sum_{b=1}^n w_b r_b$$

Thus, the model can be flexibly run using any configuration of branches, enabling our ablation study. This also enhances interpretability when using multiple branches by showing how much risk came from each input type.

We originally designed each branch to output a 64-node embedding, which could be fused using an attention mechanism into a final risk score output. However, the unmapped gene branch dominated this fusion due to its large input size and noise, yielding poor performance. Switching to the risk score late fusion described above addressed this issue. As such, the embedding fusion results are not presented. Future work should explore if attention-based embedding fusion can be done well.

### Model Training

Before model training, genomic input data was loaded and split into mapped or unmapped genes based on if they are defined in the Reactome gene-pathway mapping matrix [4]. During training, a number of techniques were applied to overcome the small dataset size and class imbalance. Data was split into 80% train and 20% test sets using an event-stratified split. The training loop consisted of event-stratified 5-fold cross validation on the training set (meaning 64% of the original data was used for training while 16% was used for validation) for a maximum of 50 epochs. If no performance improvement on the validation set is observed for 5 epochs in a row, then training on that fold will stop early to prevent overfitting. The model optimizes a Cox loss function implemented with Pycox [15]. The Adam optimizer [16] was used with an initial learning rate and weight decay of 0.001.

After cross-validation training, the model was retrained on the full training set (no validation set carved out) using the mean best number of epochs across the 5 folds. Finally, it was run on the test set for evaluation.

### Model Evaluation

The Concordance Index (C-Index) was chosen as the primary evaluation metric due to its prevalence in survival analysis [17]. It measures the proportion of the time that the model successfully orders a pair of patients in terms of their observed survival time. A C-Index of 0.5 represents random ordering, while 1.0 represents perfect ordering. Predicted risk scores for each patient and mean pathway node activations on the test set were also saved for analysis. Due to

the small dataset size, we performed an  $n=1000$  bootstrap on the test set to obtain a mean, standard deviation, and 95% confidence interval for the C-Index.

### Experimental Design

We performed an ablation study in which each model configuration (combination of the three branches) was evaluated. To account for the variability in performance caused by the small dataset size, we performed 10 experiments (random seeds 0-9) for each model configuration.

Three Cox PH models were also deployed as competitive baselines: one fit to  $n=50$  PCA features from the genes (“gene”), one fit to the 3 clinical variables (“clinical”), and a third fit to both the  $n=50$  PCA genes and the 3 clinical features (“combined”). These serve as the leading alternative model to neural network approaches.

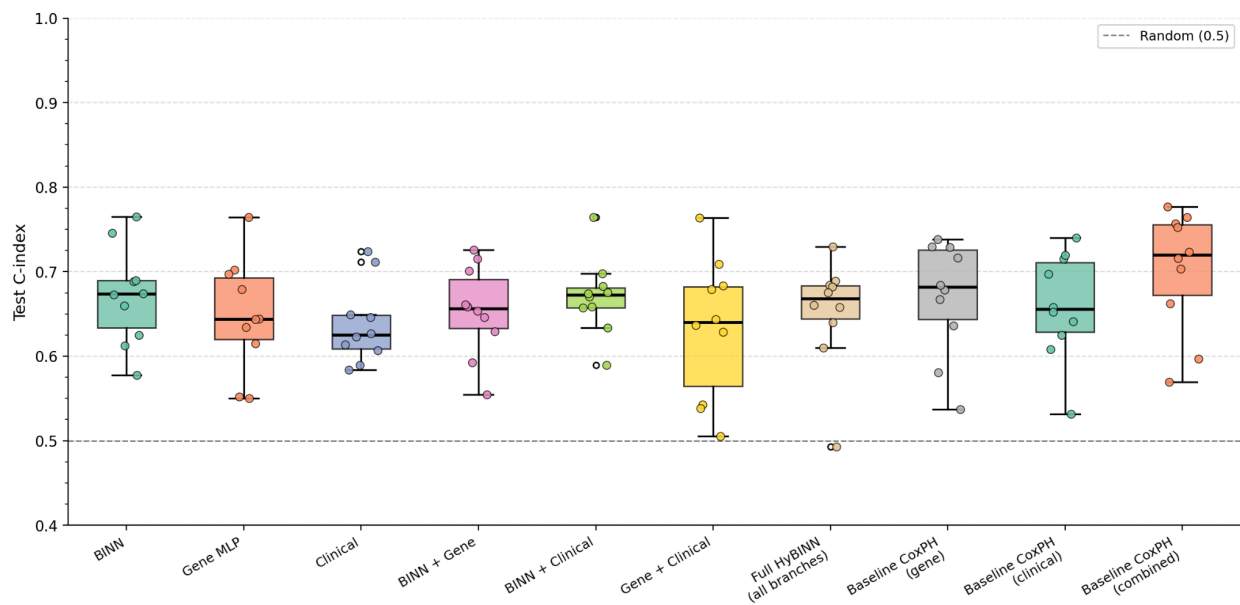
### Pathway Interpretability

To gain insight into whether the BINN branch of the model learned biologically-relevant pathways, interpretability analysis using Integrated Gradients (IG) through the Captum library [18] was performed. Pathways were required to exceed the median raw absolute IG attribution before being ranked by consistency ( $\text{mean} \div \text{SE}$ ) across seeds, analogous to a fold-change floor in differential expression, to avoid selecting for pathways that are stable simply because their contribution is uniformly negligible.

## 4. RESULTS

### Ablation Study

The results of the ablation study are shown in Figure 2 and Tables 1-2. While the distributions of all configurations overlap, qualitative differences are visible in the boxplot. Configurations including the unmapped Gene MLP have much higher standard deviation, reflecting the large input size of this branch and the noise that these genes carry. The clinical branch, on the other hand, is more robust across seeds.



**Figure 2. Ablation Study.** Test set performance (C-index) across  $n=10$  random seed experiments for each model configuration. Dotted line at C-index of 0.5, which represents random ordering.

Among HyBINN configurations, the BINN branch alone achieved the highest overall performance on the test set, with BINN + Clinical nearly as good. The clinical branch alone (no

gene expression data) is surprisingly competitive. However, the combined Cox PH model outperformed all HyBINN configurations (Table 1).

**Table 1. Ablation Study.**

| BINN Branch      | Gene Branch | Clinical Branch | Train CV C-index (mean $\pm$ std)     | Test C-index (mean $\pm$ std)         | Bootstrap 95% CI |
|------------------|-------------|-----------------|---------------------------------------|---------------------------------------|------------------|
| ✓                |             |                 | 0.6899 $\pm$ 0.0328                   | <b>0.6710 <math>\pm</math> 0.0573</b> | [0.5548, 0.7810] |
|                  | ✓           |                 | 0.6834 $\pm$ 0.0288                   | 0.6482 $\pm$ 0.0667                   | [0.5196, 0.7682] |
|                  |             | ✓               | 0.6744 $\pm$ 0.0205                   | 0.6374 $\pm$ 0.0473                   | [0.5007, 0.7596] |
| ✓                | ✓           |                 | 0.6895 $\pm$ 0.0280                   | 0.6537 $\pm$ 0.0532                   | [0.5305, 0.7662] |
| ✓                |             | ✓               | <b>0.7019 <math>\pm</math> 0.0329</b> | 0.6705 $\pm$ 0.0447                   | [0.5488, 0.7848] |
|                  | ✓           | ✓               | 0.6876 $\pm$ 0.0285                   | 0.6330 $\pm$ 0.0825                   | [0.5131, 0.7474] |
| ✓                | ✓           | ✓               | 0.6903 $\pm$ 0.0254                   | 0.6521 $\pm$ 0.0644                   | [0.5322, 0.7634] |
| CoxPH (gene PCA) |             |                 | -                                     | 0.6697 $\pm$ 0.0675                   | -                |
| CoxPH (clinical) |             |                 | -                                     | 0.6588 $\pm$ 0.0626                   | -                |
| CoxPH (combined) |             |                 | -                                     | <b>0.7021 <math>\pm</math> 0.0714</b> | -                |

N=10 experiments. Bold indicates the highest mean C-index for either HyBINN or Cox PH.

Despite qualitative differences in the ablation results, the ablation results did not demonstrate statistically significant variation across groups (Omnibus Friedman Test,  $\chi^2(9) = 12.9891$ ,  $p = 0.1631$ ,  $n=10$  paired seeds). Furthermore, none of the pairwise comparisons between configurations achieved statistical significance (Wilcoxon Signed-Rank test, Holm-corrected), and all comparisons but four yielded a Holm p-value of 1.0. Notably, all four pairwise comparisons with Holm  $p < 1.0$  involved unmapped genes, further indicating the detrimental noise that these inputs carry (Table 2).

**Table 2. Ablation Pairwise Significance.**

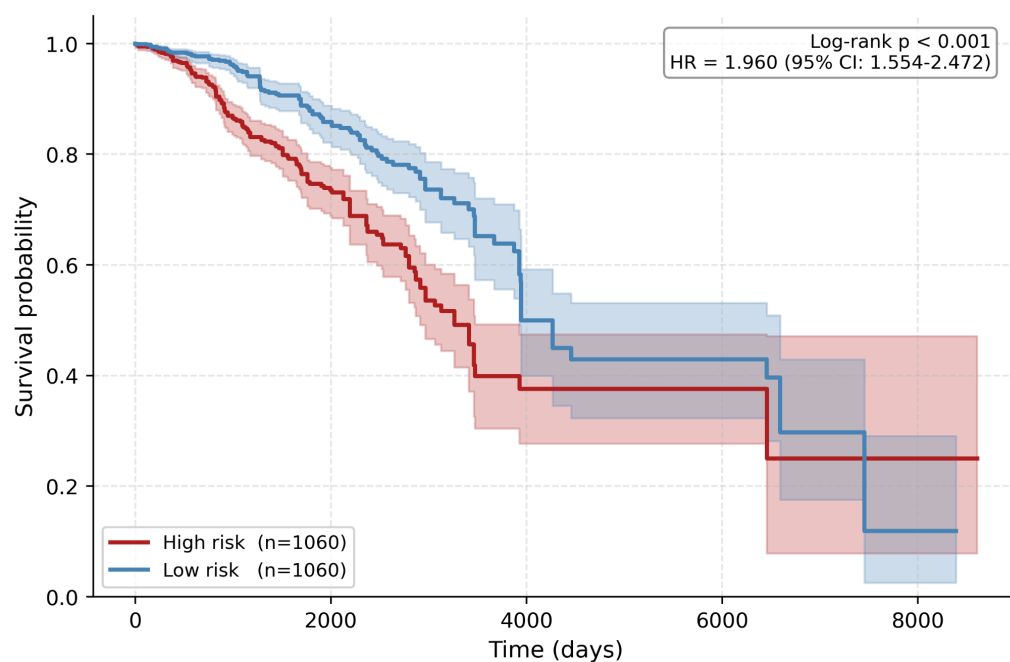
| Config A        | Config B            | Raw p-value | Holm p-value | Significant? |
|-----------------|---------------------|-------------|--------------|--------------|
| Gene            | CoxPH<br>(combined) | 0.003906    | 0.175781     | No           |
| BINN + Gene     | CoxPH<br>(combined) | 0.013672    | 0.601562     | No           |
| Gene + Clinical | CoxPH<br>(combined) | 0.019531    | 0.820312     | No           |
| CoxPH (gene)    | CoxPH<br>(combined) | 0.013672    | 0.601562     | No           |

All other configuration combinations were not significant with Holm p-value = 1.0.

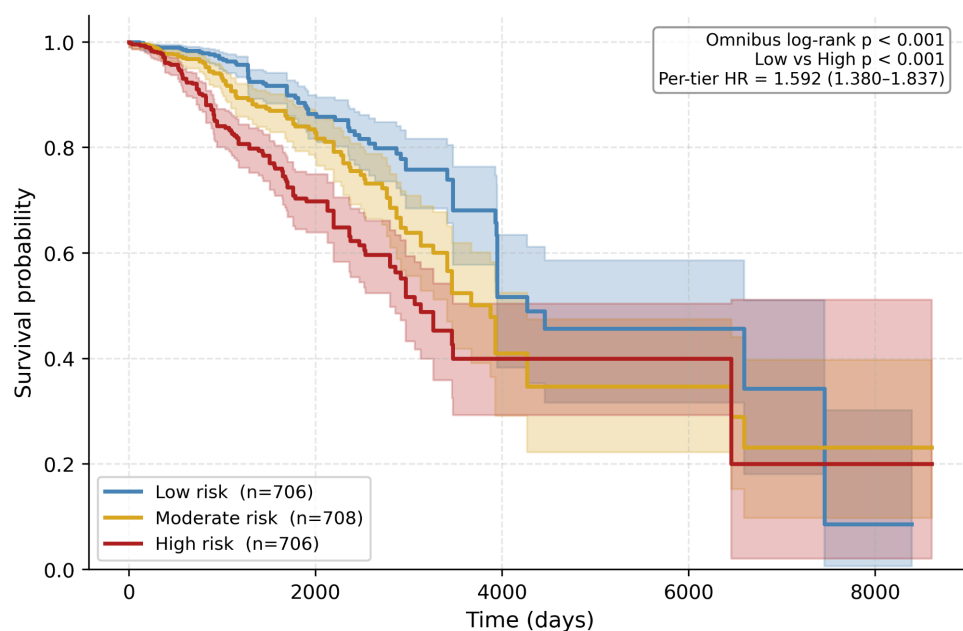
### Risk Stratification

Kaplan-Meier curves were generated to demonstrate the model's ability to stratify patients via its scalar risk score output. The test set contained only n=106 patients, so test predictions from all ten seeds were pooled for smoother curves. Note that this appears to inflate the sample size to n=2,120; however, the effective n is still 212 since these are not all unique patients. Thus, the log-rank p-values for the pooled curves aren't fully valid. Both pooled curves (Figures 3-4) and curves from a single representative seed (Figures 5-6), each with binary and tertiary splits, are shown below using the full HyBINN model's prediction.

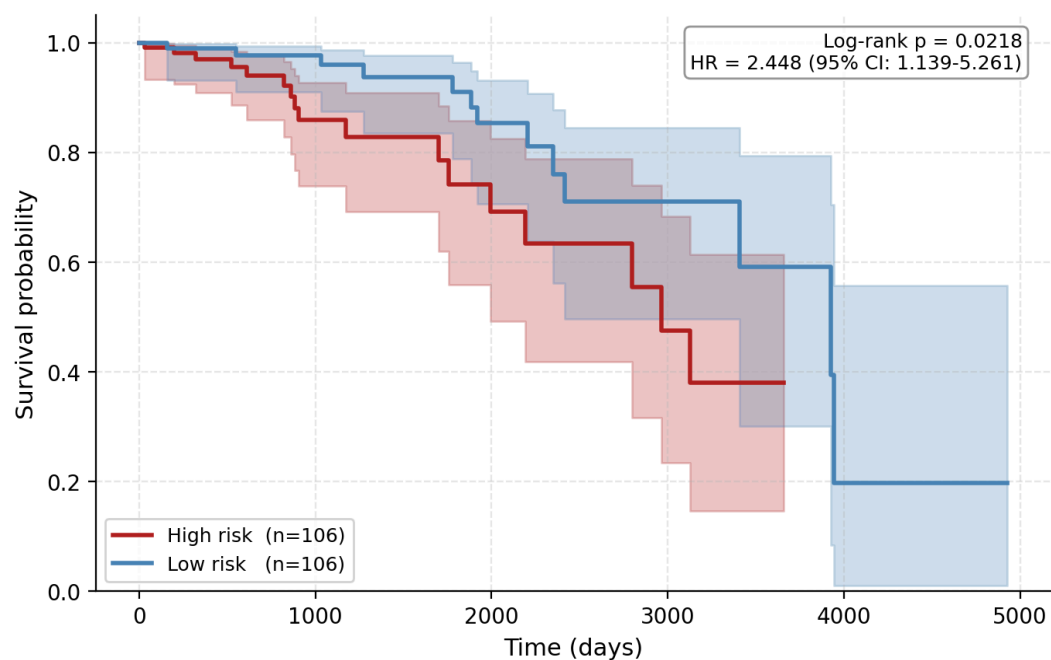
The model is able to stratify patients into two groups effectively, but three groups becomes difficult with this small of a dataset.



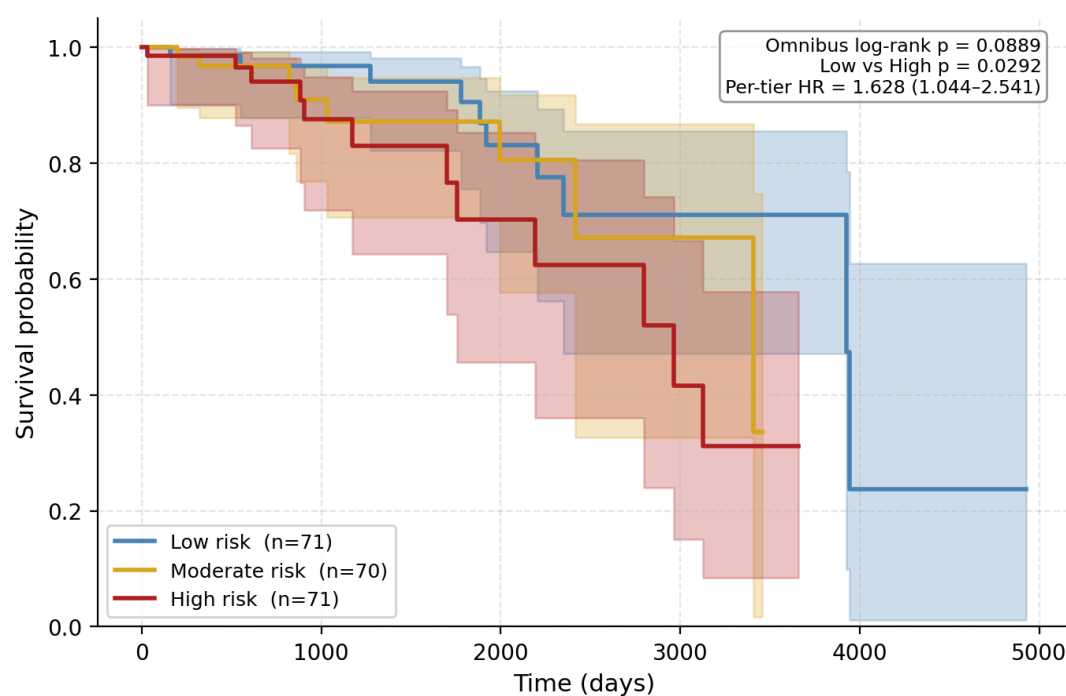
**Figure 3. HyBINN 2-Group Kaplan-Meier of All Seeds Pooled.** Patients are sorted into low or high risk by splitting on median risk score.  $n=1060$  is falsely inflated by pooling; p-value invalid.



**Figure 3. HyBINN 3-Group Kaplan-Meier of All Seeds Pooled.** Patients are sorted into groups by splitting on risk score tertiles.  $n=1060$  is falsely inflated by pooling; p-value invalid.



**Figure 5. HyBINN 2-Group Kaplan-Meier of Seed 7.** Patients are sorted into low or high risk by splitting on median risk score.



**Figure 6. HyBINN 3-Group Kaplan-Meier of Seed 7.** Patients are sorted into groups by splitting on risk score tertiles.

### Pathway Interpretability

Integrated Gradients (IG) were used to assess which biological pathways were most influential in the BINN branch of HyBINN. The mean, standard deviation (std. dev.), standard error (SE), and coefficient of variance (CV) for each pathway's IG attribution score across seeds was calculated. A consistency score was also calculated for each pathway, defined as mean attribution  $\div$  attribution standard error, representing how consistently important a pathway was across seeds.

“Top pathways” was defined as pathways with higher-than-median IG attribution score, ranked by their consistency score (higher consistency being better). The top 10 pathways according to this definition are shown in Table 3.

Eight out of the ten top pathways can be reasonably linked to breast cancer (Table 4), though not all hallmarks or canonical biomarkers are represented in this ranking.

**Table 3. Top Pathways for HyBINN Across Seeds.**

| Reactome ID    | Pathway                                                                       | Absolute IG Attribution |           |         |         | Rank by Mean IG | Consistency Score |
|----------------|-------------------------------------------------------------------------------|-------------------------|-----------|---------|---------|-----------------|-------------------|
|                |                                                                               | Mean                    | Std. Dev. | SE      | CV      |                 |                   |
| R-HSA-9 958863 | SLC-mediated transport of amino acids                                         | 0.00168                 | 0.00083   | 0.00026 | 0.49367 | 63              | 6.40563           |
| R-HSA-9 825892 | Regulation of MITF-M-dependent genes involved in cell cycle and proliferation | 0.00091                 | 0.00047   | 0.00015 | 0.51416 | 232             | 6.15035           |
| R-HSA-9 620244 | Long-term potentiation                                                        | 0.00114                 | 0.00059   | 0.00019 | 0.51546 | 160             | 6.13489           |
| R-HSA-3 89957  | Prefoldin mediated transfer of substrate to CCT/TriC                          | 0.00256                 | 0.00132   | 0.00042 | 0.51825 | 20              | 6.10189           |
| R-HSA-1 68638  | NOD1/2 Signaling Pathway                                                      | 0.00218                 | 0.00116   | 0.00037 | 0.53166 | 35              | 5.94792           |
| R-HSA-1 96807  | Nicotinate metabolism                                                         | 0.00186                 | 0.00100   | 0.00031 | 0.53521 | 54              | 5.90851           |
| R-HSA-9 861718 | Regulation of pyruvate metabolism                                             | 0.00099                 | 0.00054   | 0.00017 | 0.54130 | 201             | 5.84199           |
| R-HSA-8 955332 | Carboxyterminal post-translational modifications of tubulin                   | 0.00229                 | 0.00127   | 0.00040 | 0.55355 | 31              | 5.71275           |
| R-HSA-1 90872  | Transport of connexons to the plasma membrane                                 | 0.00268                 | 0.00155   | 0.00049 | 0.58009 | 16              | 5.45135           |
| R-HSA-9 674555 | Signaling by CSF3 (G-CSF)                                                     | 0.00099                 | 0.00058   | 0.00018 | 0.58311 | 198             | 5.42311           |

**Table 4. Biological Significance of HyBINN Top Pathways Across Seeds.**

| <b>Reactome ID</b>            | <b>Pathway</b>                                                                | <b>Relevance to Breast Cancer</b>                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">R-HSA-9958863</a> | SLC-mediated transport of amino acids                                         | Metabolic demands of cancer cells [19]                                                                                                                                |
| <a href="#">R-HSA-9825892</a> | Regulation of MITF-M-dependent genes involved in cell cycle and proliferation | n/a                                                                                                                                                                   |
| <a href="#">R-HSA-9620244</a> | Long-term potentiation                                                        | n/a                                                                                                                                                                   |
| <a href="#">R-HSA-389957</a>  | Prefoldin mediated transfer of substrate to CCT/TriC                          | Involved in tubulin folding. Tubulins form microtubules, which form mitotic spindles enabling cell division.                                                          |
| <a href="#">R-HSA-168638</a>  | NOD1/2 Signaling Pathway                                                      | Immune system and inflammation                                                                                                                                        |
| <a href="#">R-HSA-196807</a>  | Nicotinate metabolism                                                         | Precursors to NAD <sup>+</sup> and NADP <sup>+</sup> . NAD regulates glycolysis, proliferation, DNA repair, and stress response and is thus implicated in cancer [20] |
| <a href="#">R-HSA-9861718</a> | Regulation of pyruvate metabolism                                             | Expand cancer cell metabolism, enable metastasis to pulmonary tissue, protect cancer cells from oxidative stress [21]                                                 |
| <a href="#">R-HSA-8955332</a> | Carboxyterminal post-translational modifications of tubulin                   | Tubulins form microtubules, which form mitotic spindles enabling cell division.                                                                                       |
| <a href="#">R-HSA-190872</a>  | Transport of connexons to the plasma membrane                                 | Connexons are needed to form gap junctions between cells for tissue homeostasis, the loss of which is correlated with cancer development [22]                         |
| <a href="#">R-HSA-9674555</a> | Signaling by CSF3 (G-CSF)                                                     | Cytokine that promotes RAS-RAF-MEK-ERK proliferative signaling. Implicated in invasive triple-negative breast cancer [23]                                             |

## 5. DISCUSSION

### Performance

The performance of HyBINN is fair given only gene expression and clinical phenotype, but not remarkable. Overfitting is likely a core issue given the high dimension, low sample size (HDLSS) dataset, even with the sparse BINN and dropout. Other BINN models, particularly those incorporating multi-omics and imaging data, demonstrate much stronger performance for survival analysis. The main contribution of this project is not a state-of-the-art model, but rather rigorous ablation that elucidates the dynamics of gene input data on model performance.

### Implications

The ablation study, in which combinations of the BINN branch, unmapped Gene branch, and Clinical branch were used, validates the assumption that unmapped genes can be discarded for pathway-aware neural networks meant to process gene expression data. The BINN branch alone achieved the highest test C-index. The Gene branch, and combinations containing the Gene branch, yielded lower C-indices compared to those without. Additionally, the Gene branch demonstrated higher variability of outputs across seeds, indicative of noise in the unmapped gene data and the absence of the BINN's inductive bias. In other words, the BINN and clinical variables are sufficient in this HDLSS data regime; and adding more parameters (unmapped genes) does more harm than good.

Beyond the model ablation, the Cox PH models actually outperformed the neural network approaches across the board. This can be attributed to the small, sparse dataset of only RNAseq values. PCA is a strong regularization and dimensionality reduction technique, so taking the top 50 PC features captured the necessary axes of variation in the genes while providing a much

smaller input feature set for Cox PH Gene. The clinical phenotype data (T, N, M stages) were shown to carry strong predictive power implicitly for both HyBINN and Cox PH, explaining the performance of Cox PH Clinical. Finally, it makes sense that survival outcomes in this TCGA breast cancer cohort are not highly nonlinear, allowing Cox to outperform the neural networks.

While a BINN outperforms traditional neural networks in biological HDLSS settings, it appears that Cox PH can still outperform both at this sample size. Neural network approaches appear to be more useful for multi-omic and larger dataset contexts.

These findings are another piece of evidence that challenges the accuracy-interpretability tradeoff. Here, with limited HDLSS, the more interpretable models (BINN, and further, Cox PH) performed better than big neural networks.

Finally, these results also motivate expanding our biological knowledge bases such as Reactome. More mapped genes will only strengthen the performance of biologically-informed models. The more we map biology, the more we can incorporate that knowledge into predictive models grounded in and guided by science.

### Limitations

This project comes with a number of inherent limitations. The dataset is the foundational limiting factor: it represents a single cancer (breast), a single omic (RNAseq) plus clinical phenotype, and a small unbalanced cohort (n=1,059 patients after preprocessing, only n=148 of which were deceased). This is a data-sparse environment for training a neural network. It also resulted in high performance variability across runs depending on how the patients were split up, even with stratified splitting.

The multi-seed approach accounted for this variation to some degree, but also made the pathway interpretability difficult since different pathways may have been most relevant for different seeds. Each seed trained an independent model that learned on a slightly different training set with different initializations.

Additionally, Reactome contains some pathways that share many genes with other pathways. This redundancy may explain the appearance of the #2 and #3 top pathways (MITF-M-dependent genes, related to melanocytes, and Long-term potentiation, related to neurons), which themselves are not directly linked to breast cancer but may share many genes with more relevant pathways and thus were ranked high as proxies.

## 6. CONCLUSION

We developed a hybrid biologically-informed neural network (HyBINN) and performed an ablation study and comparison to traditional Cox PH models. The results indicate that unmapped genes, which are generally excluded from BINN models, are detrimental in HDLSS survival analysis settings. Furthermore, the neural network models actually failed to beat the classical baseline of a Cox Proportional Hazards model on this dataset. Nonetheless, the BINN learned biologically-meaningful pathways and achieved fair performance given a small sample size of only gene expression data. These results validate the assumptions of BINNs, but also serve as a reminder that bigger, more complex models with more features are not always better in data-sparse HDLSS contexts.

## Availability of Data and Code

The raw datasets used can be downloaded from UCSC Xena data repository, GDC TCGA Breast Cancer cohort:

[https://xenabrowser.net/datapages/?cohort=GDC%20TCGA%20Breast%20Cancer%20\(BRCA\)&removeHub=https%3A%2F%2Fxcna.treehouse.gi.ucsc.edu%3A443](https://xenabrowser.net/datapages/?cohort=GDC%20TCGA%20Breast%20Cancer%20(BRCA)&removeHub=https%3A%2F%2Fxcna.treehouse.gi.ucsc.edu%3A443). All source code, logs,

results, and figures are available in the project's GitHub repository:

<https://github.com/evanhackstadt/HyBINN/tree/main>.

## Acknowledgements

The results shown here are in whole or part based upon data generated by the TCGA Research Network: <https://www.cancer.gov/tcga>.

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